Ph.D. ENTRANCE EXAMINATION SYLLABUS

PART A: RESEARCH APTITUDE & METHODOLOGY (COMMON FOR ALL DISCIPLINES)

EXAMINATION ARCHITECTURE & ASSESSMENT OBJECTIVE:

Part A evaluates the candidate's foundational knowledge of research philosophy, scientific inquiry, analytical aptitude, formulation of research problems, hypothesis design, literature review rigor, and research publication ethics. It carries 25 Marks (Proposed) out of the total 75 Marks Ph.D. Entrance Examination.

Module / Unit	Detailed Syllabus Content & Topics
Unit 1: Fundamentals of Research	Definition, objectives, motivation, and types of research (Descriptive, Analytical, Applied, Fundamental, Quantitative, Qualitative, Conceptual, and Empirical). Scientific method and research philosophy. Criteria of good research. Research ethics and intellectual integrity.
Unit 2: Research Problem & Formulation of Hypotheses	Identification of research problem, literature review, locating research gaps, formulation of research objectives. Research hypotheses: Characteristics, types, null vs alternate hypotheses, errors in hypothesis testing (Type I and Type II errors). Feasibility and formulation of research proposal.
Unit 3: Research Design & Sampling Techniques	Meaning, need, and features of good research design. Exploratory, descriptive, and experimental research designs. Principles of experimental design. Sampling design: Census vs sample survey, probability and non-probability sampling techniques, sample size determination, sampling errors.
Unit 4: Data Collection & Measurement	Primary data and secondary data collection methods (observation, interview, questionnaires, schedules, surveys). Reliability and validity of measurement scales. Construction of research instruments and questionnaires. Pre-testing and pilot testing.
Unit 5: Statistical Analysis & Hypothesis Testing	Descriptive statistics: Measures of central tendency, dispersion, skewness, and kurtosis. Inferential statistics: Parametric tests (t-test, z-test, ANOVA/F-test) and Non-parametric tests (Chi-square test, Mann-Whitney U test, Kruskal-Wallis test). Correlation, simple and multiple regression analysis. Significance levels and p-values.
Unit 6: Research Tools & Data Interpretation	Basics of computational research tools: Statistical packages (SPSS, R, Python for Data Analysis), reference management tools (Mendeley, Zotero, EndNote), spreadsheet modeling. Data visualization techniques, graphical representation, and interpretation of empirical results.
Unit 7: Research & Publication Ethics (RPE)	Philosophy and ethics: Moral philosophy, nature of moral judgments, intellectual honesty, and research integrity. Scientific misconduct: Falsification, fabrication, and plagiarism (FFP). Redundant publications (duplicate and overlapping publications, salami slicing). UGC (Promotion of Academic Integrity and Prevention of Plagiarism in Higher Educational Institutions) Regulations. Turnitin/Urkund plagiarism detection software.
Unit 8: Technical Writing, Citations & Publishing	Structure of research thesis, research papers, and technical reports. Indexing databases: Web of Science (WoS), Scopus, PubMed, IEEE Xplore, Google Scholar. Citation indices: h-index, i10-index, impact factor (JCR), CiteScore. Standard citation and referencing styles: IEEE, APA, Harvard, Chicago, MLA. Avoiding predatory journals and publisher metrics.